```
vboxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor ./rootf
s
[sudo] password for vboxuser:
[supervisor] starting...
[supervisor] warning: cannot open /dev/container_monitor (No such file or direct
ory)
[supervisor] ready — socket at /tmp/engine.sock
```

hive

hive.c

```
vboxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
[sudo] password for vboxuser:
NAME             PID      STATE      SOFT(M)  HARD(M)
vboxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine start alpha
usage: start <name> <rootfs> <cmd>
vboxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
NAME             PID      STATE      SOFT(M)  HARD(M)
vboxuser@tutorialvm:~/OS-Jackfruit/boilerplate$
```

Conatainer running +metadata running

```
vboxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine start alpha ./root
fs /bin/sh
started 'alpha' (pid 3019)
vboxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
NAME             PID      STATE      SOFT(M)  HARD(M)
alpha            3019     running    64       128
vboxuser@tutorialvm:~/OS-Jackfruit/boilerplate$
```

Showing two containers running simulataneously is required for task 1

```
vboxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine start beta ./rootf
s /bin/sh
started 'beta' (pid 3034)
vboxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
NAME             PID      STATE      SOFT(M)  HARD(M)
alpha            3019     running    64       128
beta             3034     running    64       128
vboxuser@tutorialvm:~/OS-Jackfruit/boilerplate$
```

task 3 - CLI commandsw are working

```
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine start beta ./rootf
 /bin/sh
tarted 'beta' (pid 3034)
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
AME             PID      STATE      SOFT(M)  HARD(M)
lpha            3019     running    64       128
eta             3034     running    64       128
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine stop alpha
ent SIGTERM to 'alpha' (pid 3019)
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
AME             PID      STATE      SOFT(M)  HARD(M)
lpha            3019     running    64       128
eta             3034     running    64       128
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine logs alpha
o log yet for 'alpha'
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine start gamma ./root
s /bin/echo
tarted 'gamma' (pid 3058)
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine stop beta
ent SIGTERM to 'beta' (pid 3034)
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
AME             PID      STATE      SOFT(M)  HARD(M)
lpha            3019     running    64       128
eta             3034     running    64       128
amma            3058     stopped    64       128
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$
```

kernel module proof screenshot

```
dmesg: read kernel buffer failed: Operation not permitted
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo dmesg | tail -10
[32583.765090] systemd[1]: Starting Daily dpkg database backup service...
[32584.223950] systemd[1]: Starting Refresh fwupd metadata and update motd...
[32586.189297] systemd[1]: Starting Message of the Day...
[32586.190246] systemd[1]: Condition check resulted in Rotate log files being sk
ipped.
[32586.368796] systemd[1]: Condition check resulted in Daily man-db regeneration
 being skipped.
[32586.594866] systemd[1]: motd-news.service: Deactivated successfully.
[32586.856114] systemd[1]: Finished Message of the Day.
[32588.564995] systemd-journald[2813]: File /var/log/journal/e412c420d8aa4d6bafd
48bafe5b308d1/system.journal corrupted or uncleanly shut down, renaming and repl
acing.
[32588.964046] systemd[1]: Started Journal Service.
[34864.095030] [container_monitor] Module loaded. Device: /dev/container_monitor
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$
```

the kernel module is regesterning container=pid=3100 soft+67108864 hard+134217728

```
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine start alpha ./root
fs /bin/sh
started 'alpha' (pid 3100)
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo dmesg | tail -5
32586.856114] systemd[1]: Finished Message of the Day.
32588.564995] systemd-journald[2813]: File /var/log/journal/e412c420d8aa4d6bafd
8bafe5b308d1/system.journal corrupted or uncleanly shut down, renaming and repl
cing.
32588.964046] systemd[1]: Started Journal Service.
34864.095030] [container_monitor] Module loaded. Device: /dev/container_monitor
35049.689284] [container_monitor] Registering container= pid=3100 soft=67108864
hard=134217728
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$
```

workload baseline measurement

```
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine stop alpha
sent SIGTERM to 'alpha' (pid 3100)
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ sudo ./engine start cpu_test ./r
ootfs /bin/sh
started 'cpu_test' (pid 3119)
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ gcc -o cpu_hog cpu_hog.c
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ gcc -o memory_hog memory_hog.c
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ gcc -o io_pulse io_pulse.c
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$ time ./cpu_hog
cpu_hog alive elapsed=1 accumulator=10653517217433575490
cpu_hog alive elapsed=2 accumulator=2593753717497402870
cpu_hog alive elapsed=3 accumulator=8405154519418611781
cpu_hog alive elapsed=4 accumulator=2478192936273447891
cpu_hog alive elapsed=5 accumulator=14763193755568130918
cpu_hog alive elapsed=6 accumulator=15164315398953821445
cpu_hog alive elapsed=7 accumulator=2050242294607528079
cpu_hog alive elapsed=8 accumulator=1183526330764262545
cpu_hog alive elapsed=9 accumulator=10581971060081563295
cpu_hog done duration=10 accumulator=13437398250009249476

real    0m9.477s
user    0m6.375s
sys     0m3.093s
boxuser@tutorialvm:~/OS-Jackfruit/boilerplate$
```